

# Teachers Web Platform

## Requirements & Scope

### 1. Overview

We are looking to get an AI based web platform developed for teachers that enables them to generate structured tests and examination papers aligned with the Madhya Pradesh (MP) Board curriculum.

The platform will leverage AI-powered question generation to assist teachers in quickly creating assessment papers by selecting parameters such as medium, class, subject, chapters, question types, difficulty levels and question quantities.

The objective of this platform is to simplify the process of test or exam paper creation, while ensuring alignment with the MP Board syllabus and level.

### 2. Objectives

The primary objectives of the Teacher Web Platform are:

- Enable teachers to generate exam papers and tests efficiently
- Ensure all generated questions align with the MP Board curriculum
- Previous year questions of MP Board and NCERT questions should be included while creating the exam papers.
- Use AI to generate relevant and structured questions
- Model answers for all question papers should also be available for download by the teacher for his/her reference.
- Allow teachers to customize tests based on question types and difficulty levels
- Reduce the time required by teachers to prepare assessments manually

### 3. Access Control and Authentication

#### Teacher Login

Teachers will access the platform through phone number-based authentication.

Login process:

1. Teacher enters their registered phone number.
2. A One-Time Password (OTP) is sent to the phone number (only on WhatsApp).
3. The teacher verifies the OTP to gain access to the platform.

## **Authorized Access**

Access will be limited to pre-approved teachers only.

Only phone numbers that exist in our database will be permitted to log in.

If a phone number is not registered in the system, login will not be allowed.

## **4. Teacher Onboarding via Field Agents**

New teachers will be added to the system through our field agents.

Field agents will have access to a simple administrative interface that allows them to:

- Enter teacher's name, School details & phone number
- Submit and store these names & numbers in our private database
- Capability to enter the new number for already registered teachers

Once a teacher's phone number has been added to the system, the teacher will be able to log in using the OTP authentication mechanism automatically.

## **5. Core System Capability – AI-Based Test Generation**

The primary function of the platform is to allow teachers to generate tests or exam papers using AI assistance.

Teachers will define the structure and parameters of the test, and the system will generate questions accordingly.

The AI should ensure that generated questions are:

- Aligned with the MP Board curriculum
- Appropriate for the selected class and chapters
- Structured according to the selected question types
- Balanced according to the selected difficulty levels

## **6. Test Creation Parameters**

When generating a test paper, teachers will provide the following inputs.

### **6.1 Medium Selection**

Teachers must select the language medium.

Available options:

- Hindi
- English

The generated questions must correspond to the selected language.

## **6.2 Class Selection**

Teachers must select the class for which the test is to be created.

Supported classes will correspond to the MP Board curriculum.

## **6.3 Subject Selection**

Teachers will select the subject for which the test is to be generated.

## **6.4 Chapter Selection**

Teachers will select one or multiple chapters from which the test should be generated.

This enables teachers to create:

- Chapter-wise tests
- Multi-chapter tests / papers
- Revision tests of whole syllabus

## **6.5 Exam type Selection**

Teachers will be able to select the type of examination for which they want to create the papers.

This can include:

- Based on selected chapters
- Half Yearly
- Preboards / Final Exams (Full Syllabus)

If they select any of these options they can bypass the next step of selecting the type of questions.

Other exams can also be:

- Unit Tests
- Class Tests

For any of these, teachers will have to select the type and quantity of questions in next steps.

## **6.6 Question Type Selection**

Teachers will select the types of questions to include in the test.

Supported question formats will include:

- Multiple Choice Questions (MCQs)
- True / False
- Fill in the Blanks
- Match the Following
- 1 Mark Questions (Subjective)
- 2 Mark Questions (Subjective)
- 3 & 4 Mark Questions (Subjective)

Teachers will also specify the number of questions required for each type.

### **6.7 Difficulty Level Distribution**

Teachers will be able to specify the difficulty level distribution of the questions.

Difficulty levels will include:

- Easy
- Medium
- Hard

Or

- MP Board Level (it will help teachers even more)
- A slider can be given for difficulty levels, if one is increased others automatically decrease simultaneously with it in proportion

Teachers can define how many questions should be generated for each difficulty level or they can use the slider to automatically adjust the difficulty level by just moving one of the levels.

This will allow teachers to generate balanced assessment papers suitable for classroom evaluation.

### **7. AI Question Generation**

Based on the selected parameters, the platform will generate questions using AI

The AI should ensure that:

- Questions are academically accurate
- Questions match the selected difficulty level
- Questions are relevant to the selected chapters
- Questions match the selected question format

### **8. Generated Output**

After the teacher submits the test parameters, the system will generate a complete test paper.

The output should include:

### **Question Paper**

A structured paper containing all generated questions organized by question type as per the MP Board syllabus and blueprint.

### **Answer Key / Solutions**

The teacher should be able to generate and download the complete answer key for the generated paper.

### **Answer Key / Solutions**

The generated paper should also include:

- Correct answers for objective questions
- Answer keys or solutions for descriptive questions

## **9. Output Format**

Teachers should be able to:

- Preview the generated paper
- Download the paper in a print-ready format (pdf or doc)

## **10. Data Alignment and Academic Accuracy**

All generated questions must remain aligned with:

- MP Board syllabus and level
- Selected class and subject
- Selected chapters

This ensures the generated assessments remain academically relevant and curriculum compliant.

## **11. Future Expansion (Optional Enhancements)**

Potential future enhancements may include:

- Teacher editing of generated questions
- Saving and managing previously generated test papers
- AI-assisted lesson planning

## **12. Add questions/data – For Publisher**

- An AI-assisted bulk upload option is needed where questions can be uploaded from our book PDFs to the database.

## **13. Transaction trail of usage of the Web Platform**

- The history of usage by all users should be maintained in the system for future reference by the teachers and Shivalal.
- Saving and managing previously generated test papers.

## **14. Storage of all questions in a database**

- All new/unique questions generated (with answers) for the tests should also be stored in a database for future use.

## **15. Future Expansion**

Potential future enhancements may include:

- Online manual evaluation of answer sheets.
- Evaluation of answer sheets using AI.
- AI-assisted lesson planning.

# Development Roadmap

## 1. Access & Authentication

- Phone number login system
- OTP verification via WhatsApp
- Allow login **only for phone numbers present in database**
- Reject login for unregistered numbers

## 2. Teacher Onboarding System

- Admin interface for **field agents**
- Field agents can:
  - Add teacher name
  - Add school details
  - Add phone number
- Store teacher records in **private database**
- Allow adding **additional phone numbers for existing teachers**

## 3. Test Creation Interface

Build teacher dashboard where teachers can create tests by selecting parameters.

### Medium Selection

- Hindi
- English

### Class Selection

### Subject Selection

### Chapter Selection

- Single chapter
- Multiple chapters
- Whole syllabus selection

## 4. Exam Type Selection

Teachers choose exam type:

### Major Exams

- Half Yearly
- Preboards

- Board / Final Exam

*(Selecting these bypasses question-type selection)*

#### **Other Tests**

- Unit Tests
- Class Tests

*(For these, teachers must define question types and quantities)*

### **5. Question Structure Setup**

#### **Question Type Selection**

Support following formats:

- MCQs
- True / False
- Fill in the Blanks
- Match the Following
- 1 Mark Questions
- 2 Mark Questions
- 3 & 4 Mark Questions

Teachers define number of questions for each type.

#### **Difficulty Level Distribution**

Provide difficulty configuration:

Options:

- Easy
- Medium
- Hard

or

- MP Board Level

Difficulty adjustment:

- Slider to balance difficulty levels automatically
- Changing one level adjusts others proportionally

### **6. AI Question Generation**

AI generates questions based on:



- Medium
- Class
- Subject
- Selected chapters
- Question type
- Difficulty distribution

AI must ensure:

- Academic accuracy
- MP Board alignment
- Correct question format
- Chapter relevance

## 7. Test Paper Generation

System generates:

### Question Paper

- Structured paper organized by question type

### Answer Key / Solutions

- Correct answers for objective questions
- Solutions / keys for descriptive questions

## 8. Output & Export

Teachers can:

- Preview generated paper
- Download paper in **print-ready format**
  - PDF
  - DOC

## 9. Future Enhancements

- Teacher editing of generated questions
- Save and manage previously generated papers
- AI-assisted lesson planning (optional)